Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION December-2025

NAME OF THE COURSE : M.Tech (Structural Engineering)
NAME OF THE PAPER : **Theory of Elasticity and Plasticity**
TIME: 3 Hours

SEMESTER : I
PAPER CODE : MSE 101
MAX MARKS : 100

Note:-

i. All questions are compulsory.
ii. Answer any two sub questions among a, b & c in each main question.
iii. Total Marks in each main are twenty. Each Question carries 10 Marks.

Q1 (2X10=20 marks)

(a) What is generalized Hooke's law? Establish the stress – strain relationship for isotropic materials and hence the relationship between E, G and K. (CO 1)

(b) The state of stress at a point is given by following tensor. Determine the principal ptresses and their directions? (CO 1)

$$\begin{bmatrix} 9 & 8 & 6 \\ 8 & 5 & 2 \\ 6 & 2 & 4 \end{bmatrix}$$

(c) The stress components at a point in a body are given as following. Determine wether these stress components satisfies the equilibrium equations or not at a point (1,-1,2). If not find the body force required at that point to keep the body in equilibrium? (CO 1)

$$\sigma x=3xy^2z+2x,\ \sigma y=5xyz+3y,\ \sigma z=x^2y+y^2z,\ \tau xy=0,\ \tau yz=\tau xz=3xy^2z+2xy.$$

Q2 (2X10=20 marks)

(a) Define principal stresses and principal directions. Show that the determination of principal stresses and principal directions reduces to the solution of an Eigen value problem? (CO 2)

(b) A metal sheet is deformed uniformly in its own plane with a set of strain components as $\varepsilon_x=-200X\ 10^{-6}$, $\varepsilon_y=800X10^{-6}$, $\gamma_{xy}=-500X\ 10^{-6}$. find the strain components associated with a transformed coordinate axis by 30 degrees clockwise to the original coordinate Axes? (CO 2)

(c) Proove that $9\tau^2_{oct} = (\sigma_1 - \sigma_2)^2+(\sigma_2 - \sigma_3)^2+(\sigma_3 - \sigma_1)^2$? (CO 2)

Q3 (2X10=20 marks)

(a) Explain the plane stress and plane strain problems? (CO 3)

(b) Derive equilibrium equations in polar coordinate for a plane strain system? (CO 3)

(c) The stress tensor at a point is given as follows. Determine the strain tensor at this point? (CO 3)

Take E=210X 10^6 kN/m2, v=0.3 ? $\begin{bmatrix} 200 & 160 & -120 \\ 160 & -240 & 100 \\ -120 & 100 & 160 \end{bmatrix}$ kN/m^2

Q4 (2X10=20 marks)

(a) Derive the transformation equations for 3D stress state? (CO 4)

(b) Prove that elasticity solution to a beam subjected to pure bending will give the same result of theory of simple bending? (CO 4)

(c) What are the strain invariants while calculating the principal strain in 3-dimensions, Calculate the principal strains for the given strain matrix? (CO 4)

$$\varepsilon = 10^{-4} \begin{pmatrix} 2 & 6 & 2 \\ 6 & 9 & 5 \\ 2 & 5 & 6 \end{pmatrix}$$

Q5 (2X10=20 marks)

(a) Write short notes on Plastic Failure theories, yield conditions. (CO 5)

(b) Assuming zero body forces derive the compatibility equation in terms of stress functions for plane stress problem? (CO 5)

(c) Explain the Treska and Mohrs strength theories? (CO 5)